

Activity 15.07.2026

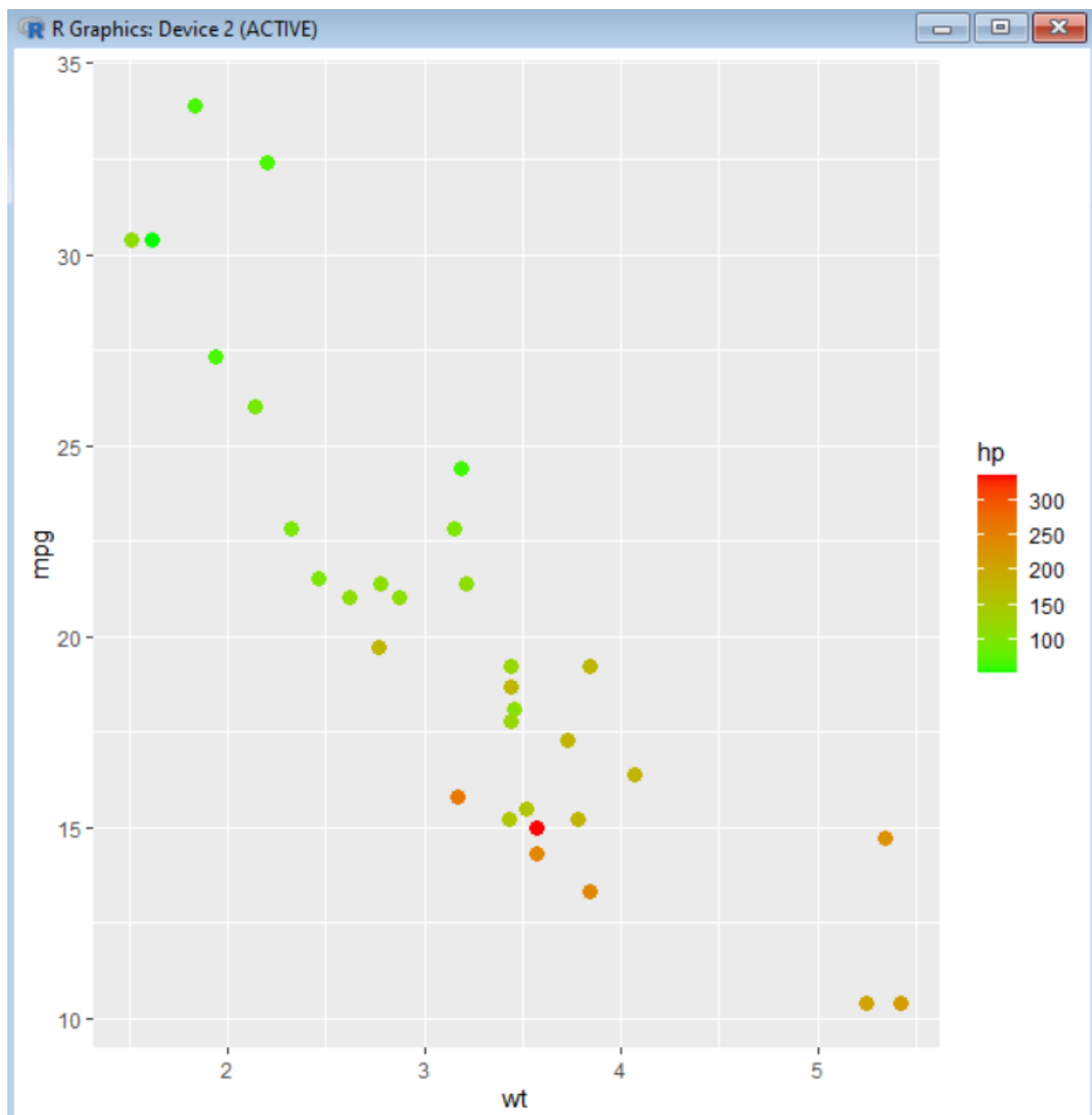
1.Smart City Air Quality Monitoring

Library(ggplot2)

```
ggplot(mtcars, aes(wt, mpg, color = hp)) +
```

```
  geom_point(size = 3) +
```

```
  scale_color_gradient(low = "green", high = "red")
```



2. Earthquake Risk Analysis:

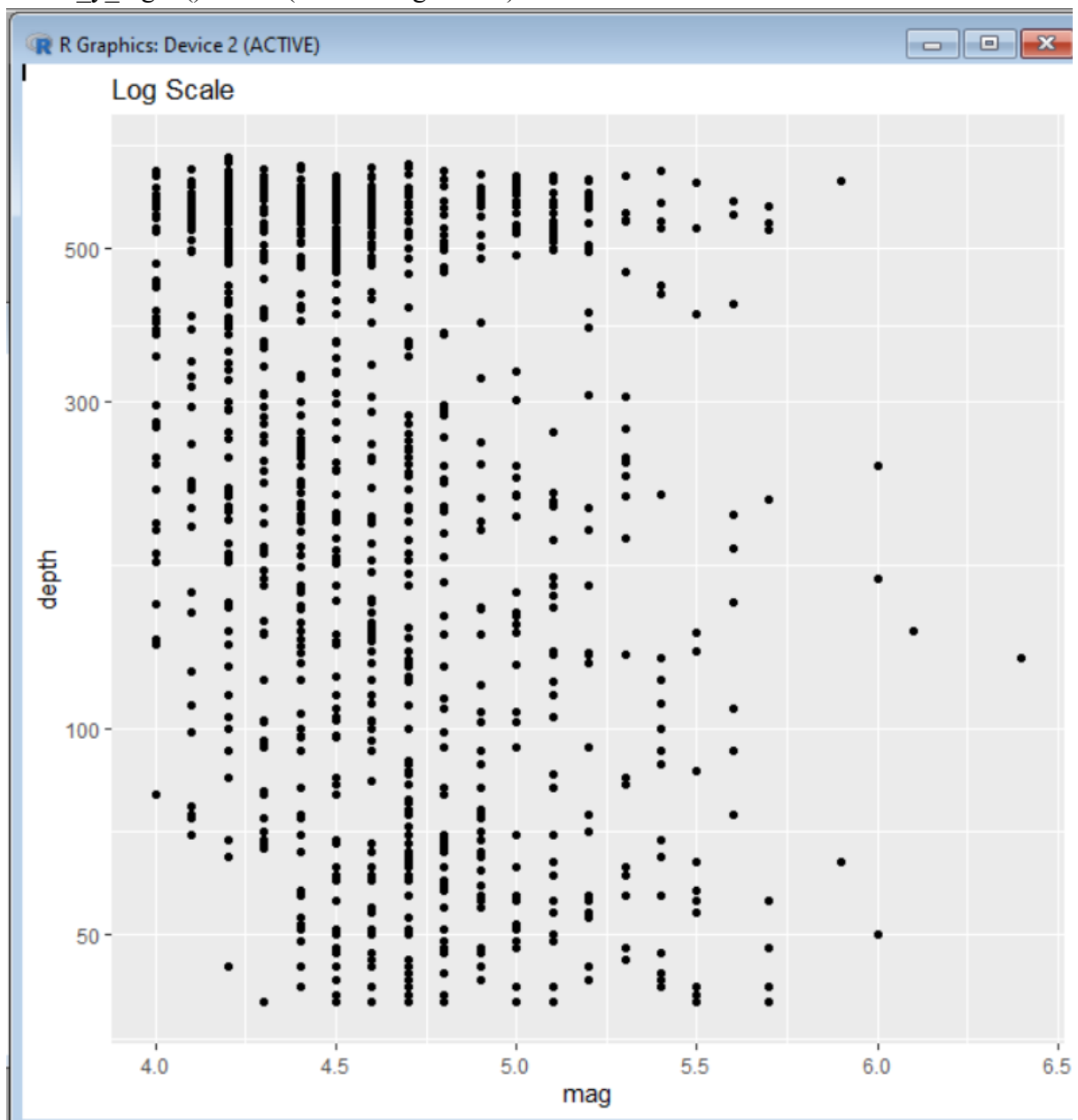
Library(ggplot2)

```
ggplot(quakes, aes(mag, depth)) + geom_point() +
```

```
  labs(title = "Linear Scale")
```

```
ggplot(quakes, aes(mag, depth)) + geom_point() +
```

```
  scale_y_log10() + labs(title = "Log Scale")
```



3. Wind Direction Analysis for Airport Operations:

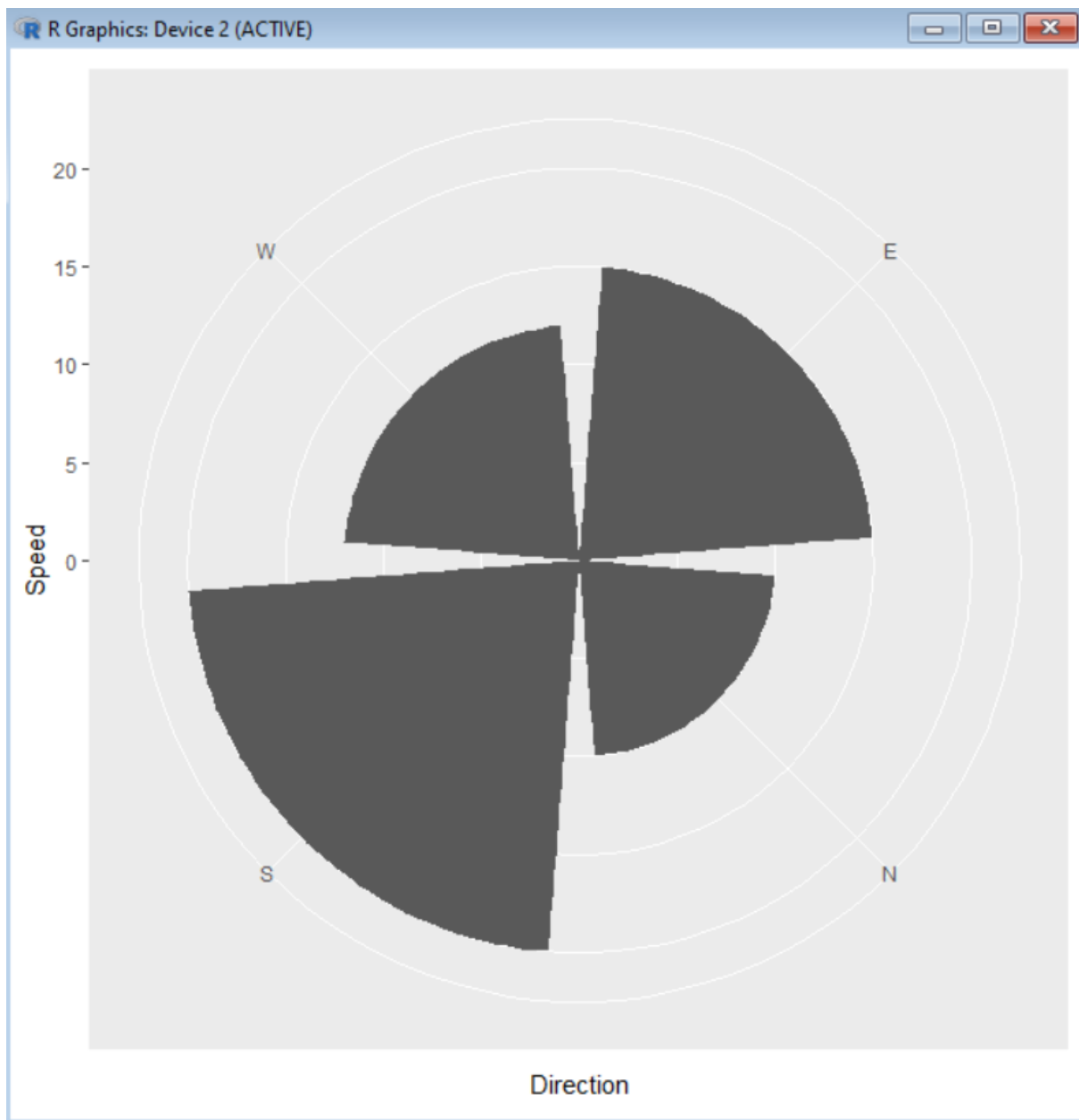
```
library(ggplot2)
```

```
wind <- data.frame(
```

```
  Direction = c("N", "E", "S", "W"), Speed = c(10, 15, 20, 12))
```

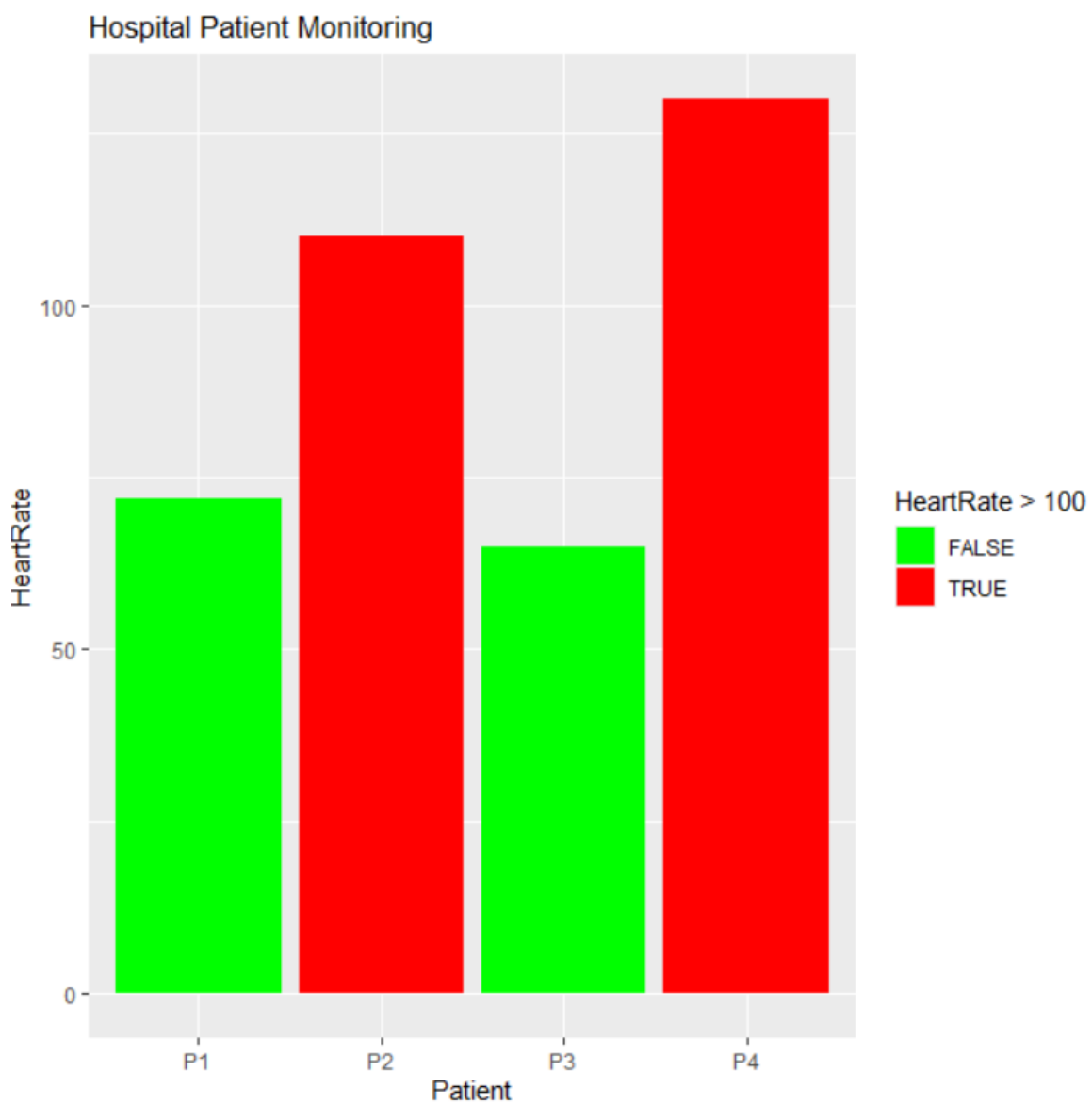
```
ggplot(wind, aes(Direction, Speed)) + geom_bar(stat = "identity")
```

```
ggplot(wind, aes(Direction, Speed)) + geom_bar(stat = "identity") + coord_polar()
```



4. Hospital Patient Monitoring Dashboard:

```
patient <- data.frame(  
  Patient = c("P1", "P2", "P3", "P4"), HeartRate = c(72, 110, 65, 130))  
ggplot(patient, aes(Patient, HeartRate, fill = HeartRate > 100)) + geom_bar(stat = "identity")  
+  
  scale_fill_manual(values = c("green", "red")) +  
  labs(title = "Hospital Patient Monitoring")
```



5.Global Climate Change Study:

```
library(ggplot2)
```

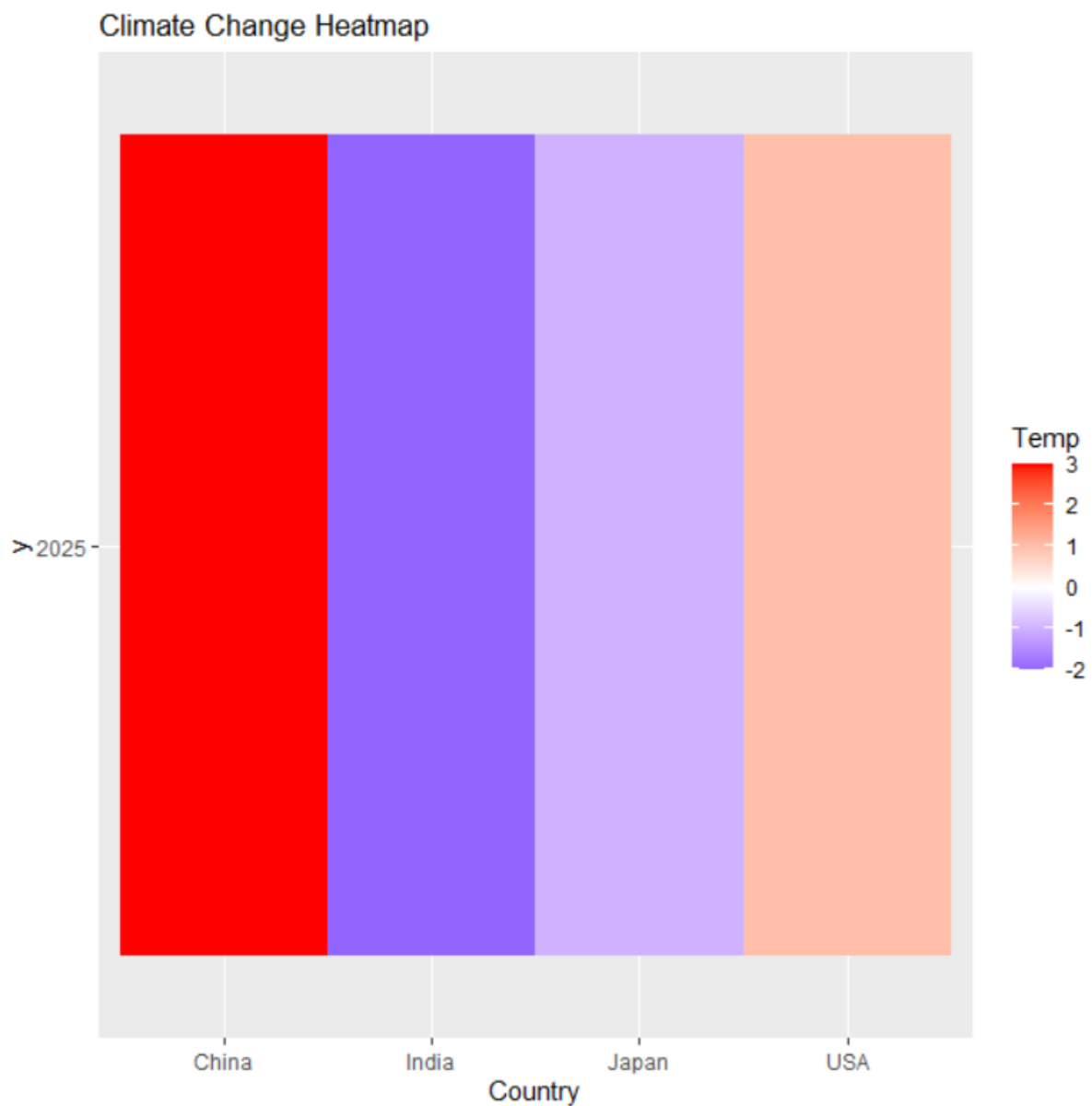
```
data <- data.frame(
```

```
  Country = c("India","USA","China","Japan"),Temp = c(-2, 1, 3, -1))
```

```
ggplot(data, aes(x = Country, y = "2025", fill = Temp)) +geom_tile() +
```

```
scale_fill_gradient2(low = "blue", mid = "white", high = "red") +
```

```
labs(title = "Climate Change Heatmap")
```



6.E-Commerce Sales Analytics:

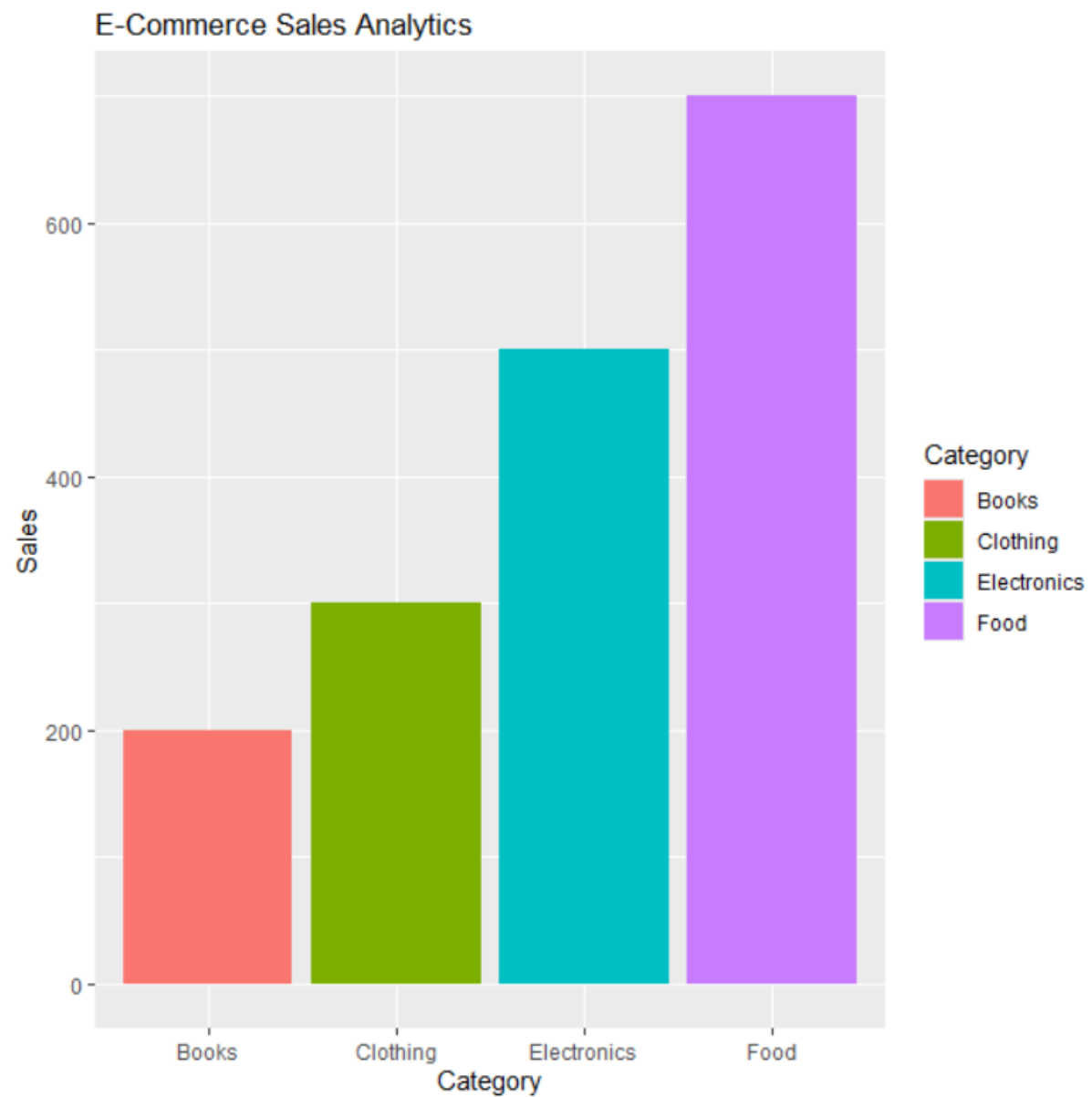
```
library(ggplot2)
```

```
sales <- data.frame(
```

```
  Category = c("Electronics","Clothing","Food","Books"), Sales = c(500,300,700,200))
```

```
ggplot(sales, aes(x = Category, y = Sales, fill = Category)) + geom_col() +
```

```
labs(title = "E-Commerce Sales Analytics")
```



7.Satellite Image Vegetation Analysis:

```
library(ggplot2)
```

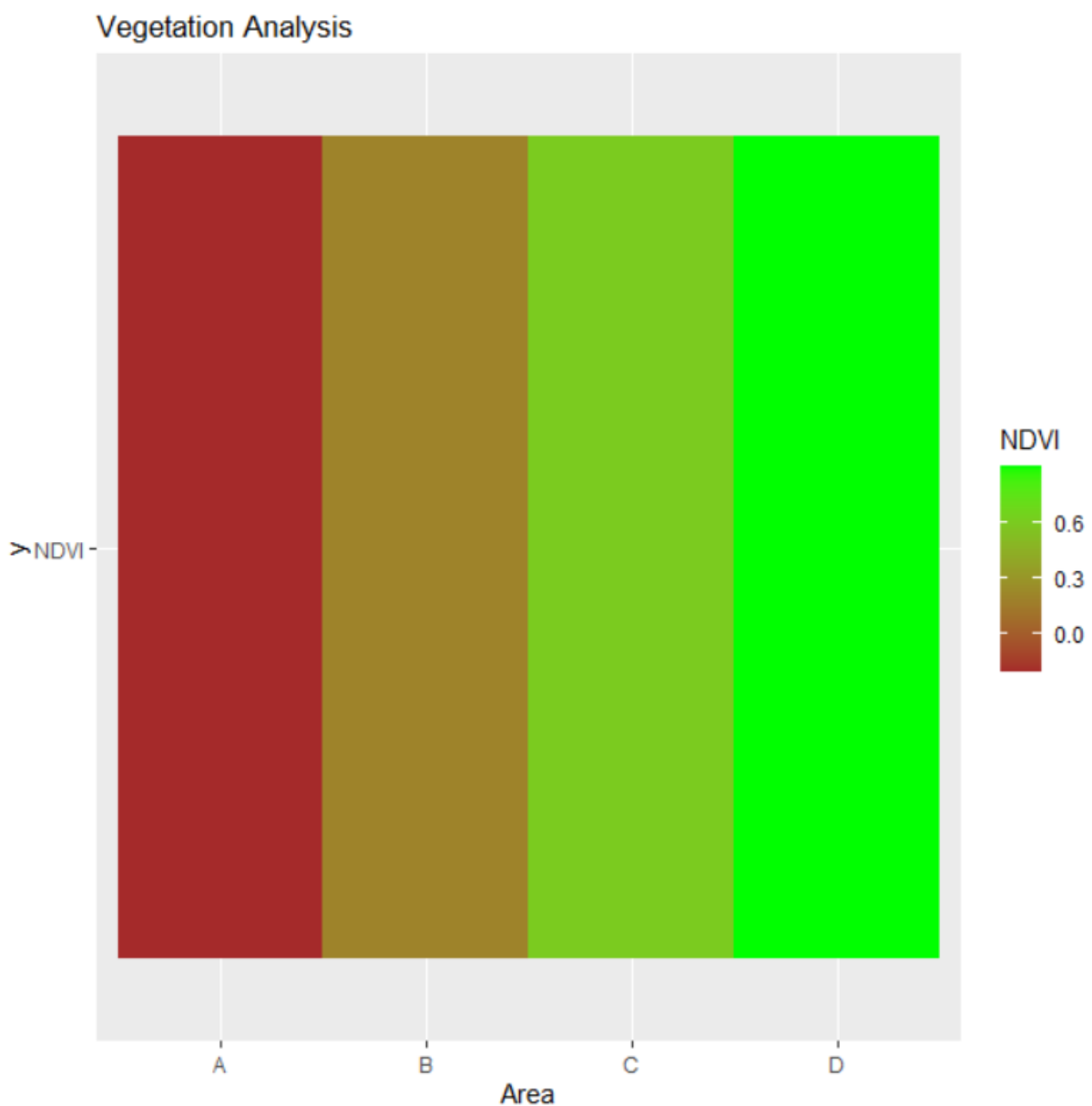
```
data <- data.frame(
```

```
  Area = c("A","B","C","D"), NDVI = c(-0.2, 0.2, 0.6, 0.9))
```

```
ggplot(data, aes(x = Area, y = "NDVI", fill = NDVI)) +geom_tile() +
```

```
  scale_fill_gradient(low = "brown", high = "green") +
```

```
  labs(title = "Vegetation Analysis")
```



8.Cryptocurrency Market Volatility:

```
library(ggplot2)
```

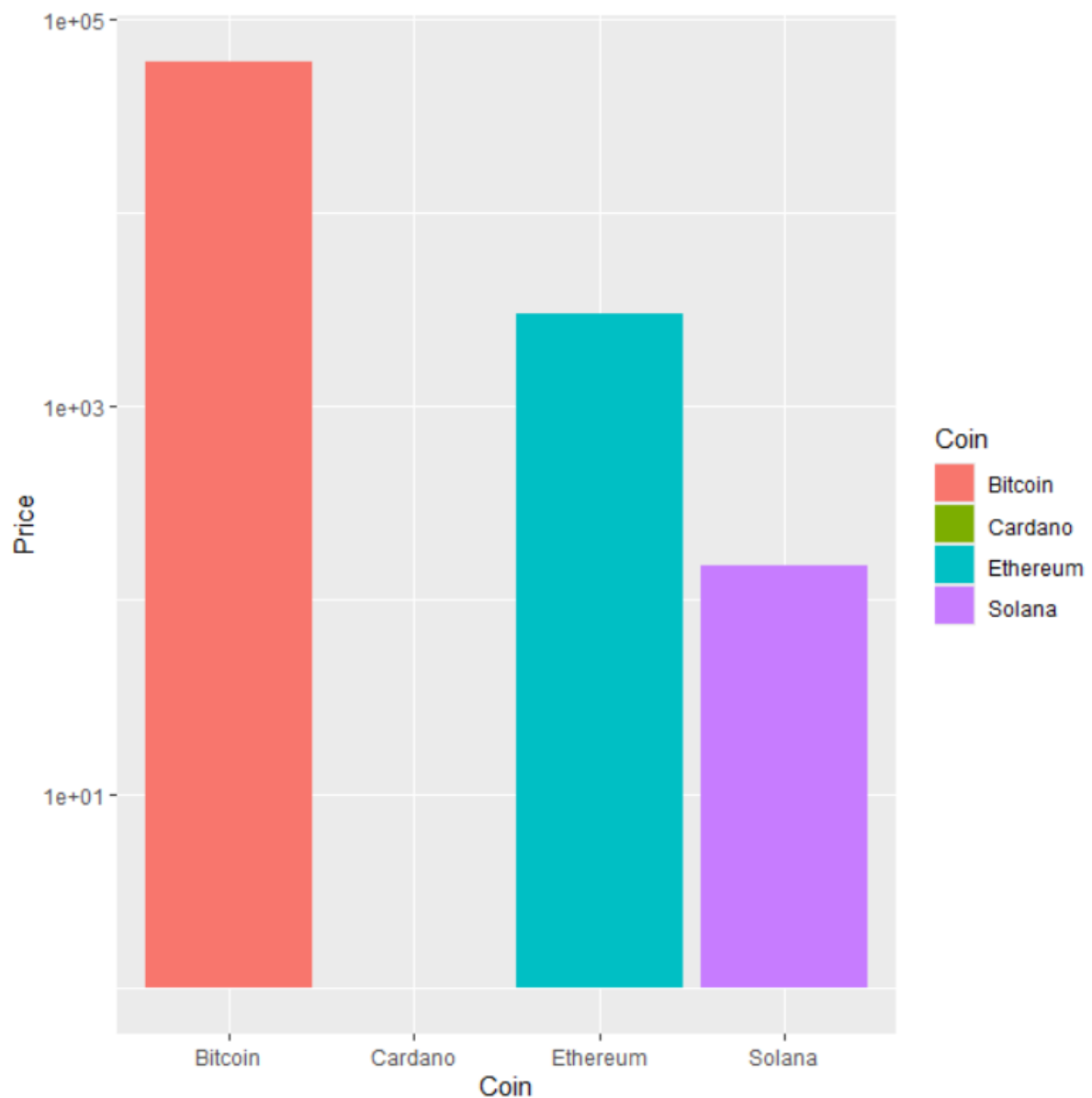
```
data <- data.frame(
```

```
  Coin = c("Bitcoin","Ethereum","Solana","Cardano"),
```

```
  Price = c(60000,3000,150,1))
```

```
ggplot(data, aes(Coin, Price, fill = Coin)) +geom_col()
```

```
ggplot(data, aes(Coin, Price, fill = Coin)) +geom_col() +scale_y_log10()
```



9.Marathon Runner Performance Analysis:

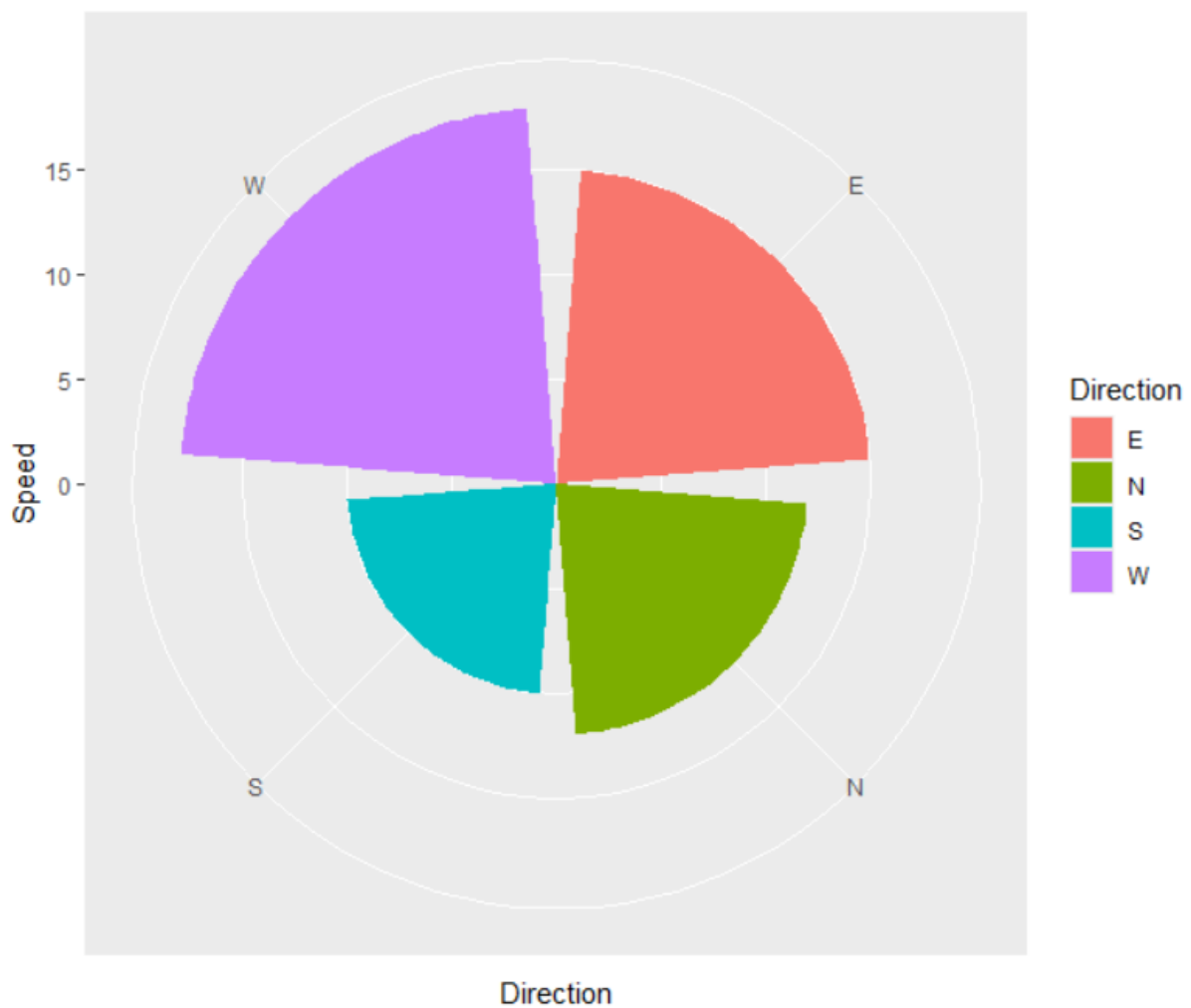
```
library(ggplot2)
```

```
runner <- data.frame(
```

```
  Direction = c("N","E","S","W"),Speed = c(12,15,10,18))
```

```
ggplot(runner, aes(Direction, Speed, fill = Direction)) + geom_col()
```

```
ggplot(runner, aes(Direction, Speed, fill = Direction)) +geom_col() +coord_polar()
```



10.COVID-19 Vaccination Campaign Dashboard:

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  District = c("A","B","C","D"),Rate = c(60,75,85,95))
```

```
ggplot(data, aes(District, Rate, fill = Rate)) +geom_col() +
```

```
  scale_fill_gradient(low = "yellow", high = "green")
```

